# Do Large Language Models Implicitly Encode Deontic Reasoning?

Exploring Faithfulness to Formal Deontic Logic

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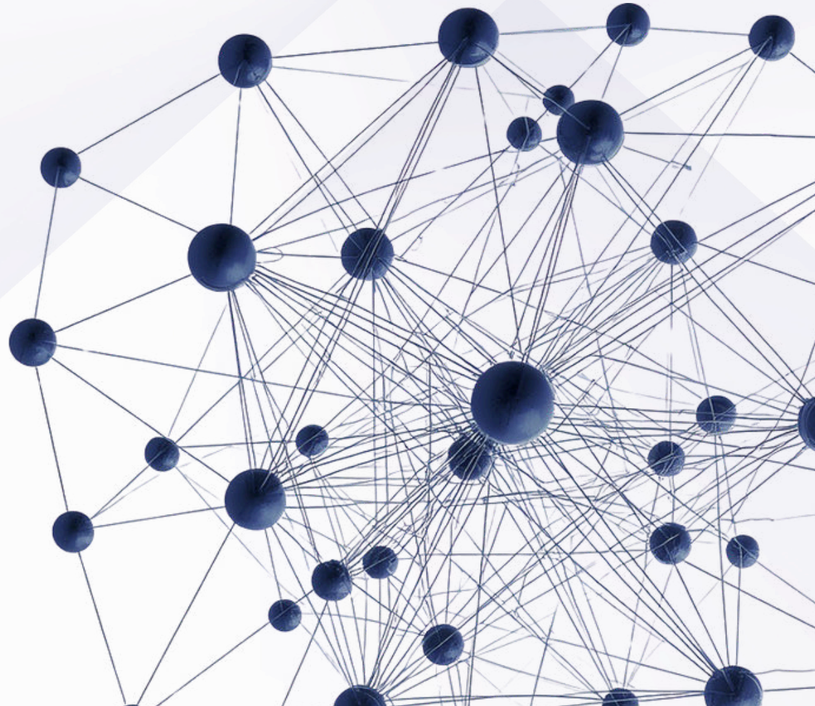


# INTRODUCTION & SCOPE

Deontic logic studies obligations and permissions
("deon" = duty in Greek).

“You may leave" → true in some deontic worlds (permission)
"You must leave" → true in all deontic worlds (obligation).

Relevance to AI:
Do LLM consistenty exhibit rule usage, or dependent on form?



# CORE TERMINOLOGY IN DEONTIC LOGIC

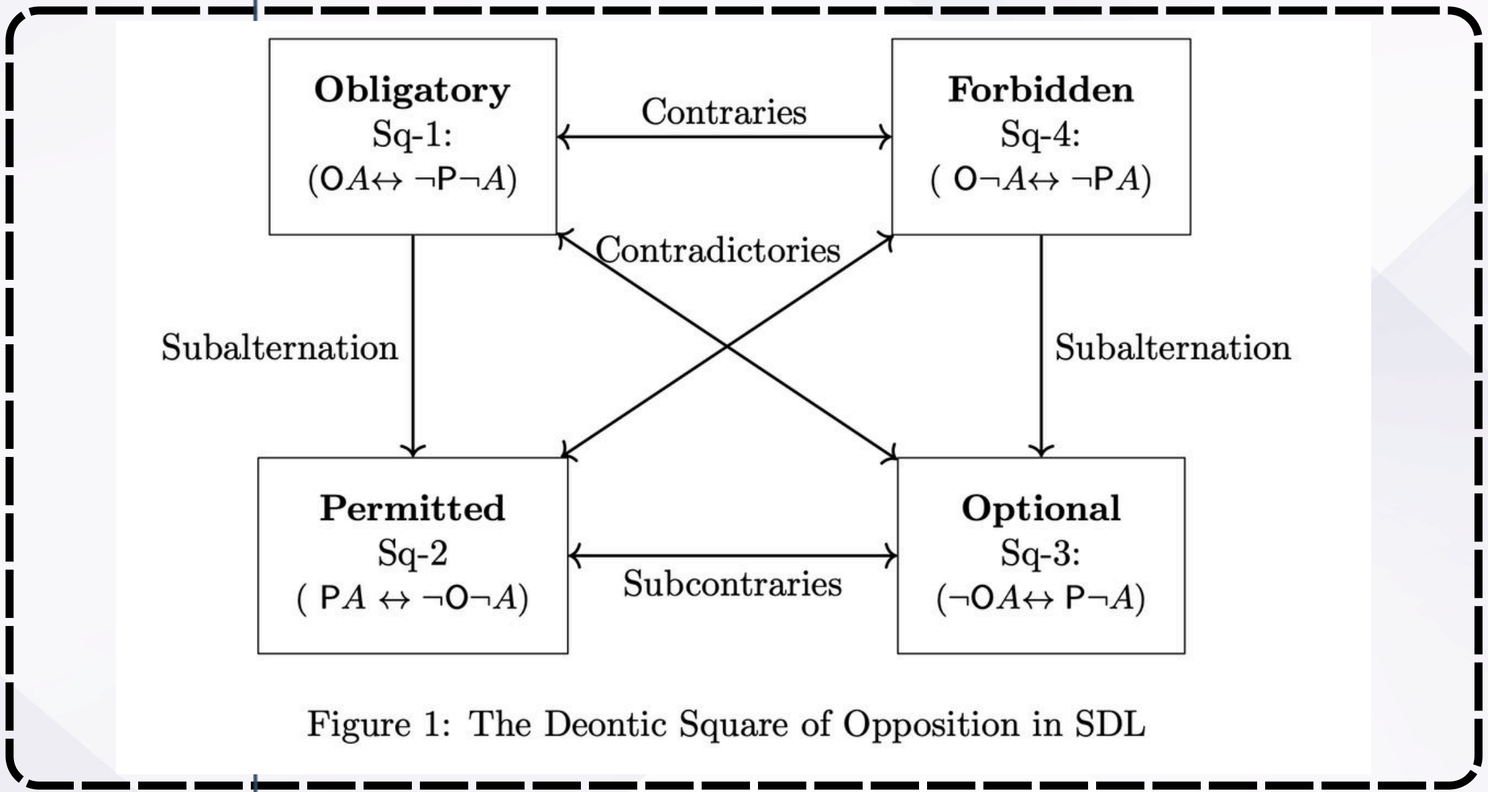



Figure 1: The Deontic Square of Opposition in SDL

# RESEARCH QUESTIONS

RQ1: Which syntactic forms cause LLM failures in deontic reasoning tasks?

RQ2: Can LLMs recover semantic verdicts from explicit Kripke-style scenarios using formal structure rather than world knowledge?

RQ3: Which deontic proof rules do LLMs consistently follow or systematically violate?

RQ4: Are observed failure patterns consistent with known biases in language model reasoning?



# BENCHMARK METHODOLOGY

GPT-4 | Claude | Deepseek V4 | Gemini

Models evaluated

## Proof layer

Tests adherence to SDL axioms & deontic square

- **Inference rule tasks** — SDL axiom adherence
- **Deontic square** — O / P / F / PF relations
- **Failure patterns** — All 4 models analysed

## Semantic layer

Eliminates world knowledge bias

- **Kripke scenarios** — Possible-worlds tasks
- **Nonsense predicates** — Strips statistical shortcuts
- **Formal vs. statistical** — Reasoning source probe

## Syntax layer

Isolates syntactic processing capabilities

- **Controlled fragments** — Linguistic variation testing
- **Negation scope** — Boundary disambiguation
- **Operator inversions** — Templates & permutations

# SDL AXIOMS

N (Necessitation): If $\vdash \varphi$, then $\vdash O\varphi$ — if something is a theorem, it is obligatory
K (Distribution): $O(\varphi \rightarrow \psi) \rightarrow (O\varphi \rightarrow O\psi)$ — obligation distributes over implication

D (Consistency): $OA \rightarrow PA$ — what is obligatory must be permissible
Definition: $PA \equiv \neg O \neg A$ — permission means the negation is not obligatory



# KEY TEST RESULTS – SYNTACTIC AND SEMANTIC LAYERS

## Paraphrase Consistency (S1): 0% failure rate

vocabulary variations and synonymous phrasing do not confuse models, demonstrating robust lexical flexibility in deontic reasoning tasks.

## Negation Scope (S2): 0% failure rate

models reliably distinguish between absence of obligation versus explicit prohibition.

## Operator Inversion (S3): Mostly correct responses;

notably, Deepseek V4 shows significant improvement when using Chain-of-Thought prompting.



# KEY TEST RESULTS – CHALLENGING PARADOXES (S4 & S5)

**Ross's Paradox & Free Choice (S4)**
All models near-perfect. Models follow formal SDL logic even when it defies intuition.
$OA \rightarrow O(A \vee B)$
"If you must post the letter, must you post it or burn it?"

**Contrary-to-Duty Paradoxes (S5)**
Significant predicted failures. When obligations conflict, models prefer natural language coherence over formal consistency.
“you ought to keep your promise, but if you break it you ought to apologise.”



# SEMANTIC LAYER TESTING WITH KRIPKE SCENARIOS

Testing formal semantic understanding through controlled scenarios:

- Nonsense predicates
- LLMs reason from formal logics OR on statistical patterns
- Kripke-style scenarios

**’if the blewble was blopped, and the flig was zandered’ together with ’the flig was not zandered, can we infer ’the flig was not blopped’ ?**

# SUMMARY OF KEY FINDINGS

**Models handle surface-level deontic language well**
vocabulary changes and negation ("not allowed" vs. "forbidden") don't trip them up.

**Negation is handled correctly**
models reliably distinguish "not obligated" from "forbidden," with no failures recorded.

**Whether deeper logic exposes further failures is still under investigation**
semantic reasoning across Kripke scenarios and proof-layer consistency under chained obligations remain to be tested.

**The pattern matching hypothesis is yet to be confirmed**
whether models rely on statistical shortcuts rather than formal reasoning will be examined in the semantic and proof layers.



# IMPLICATIONS, LIMITATIONS, AND FUTURE DIRECTIONS

Implications:
• Deontic reasoning is crucial for AI systems handling norms, rules, and ethics
• Mixed capabilities in current LLMs highlight significant gaps in formal logical understanding
• Models may produce plausible-sounding but logically inconsistent normative judgments

Limitations & Future Directions:
• Rapidly evolving LLM landscape may change results over time
• Difficulty isolating effects of architecture vs training vs prompt engineering
• Future work: mechanistic analyses of failure causes and continuous evaluation as models develop

# THANK YOU

Questions and Discussion
Welcome

